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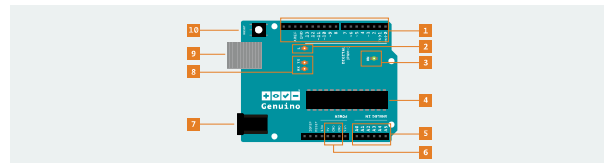
UNO Board Anatomy

Arduino UNO Board Anatomy

An overview of the classic Arduino UNO.

Author • Arduino Last revision • 10/07/2025

Arduino boards sense the environment by receiving inputs from many sensors, and affect their surroundings by controlling lights, motors, and other actuators. Arduino boards are the microcontroller development platform that will be at the heart of your projects. When making something you will be building the circuits and interfaces for interaction, and telling the microcontroller how to interface with other components. Here the anatomy of Arduino UNO.



- 1 **Digital pins** Use these pins with `digitalRead()`, `digitalWrite()`, and `analogWrite()`.

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- 2 **Pin 13 LED** The only actuator built-in to your board. Besides being a handy target for your first blink sketch, this LED is very useful for debugging.
- 3 **Power LED** Indicates that your Arduino is receiving power. Useful for debugging.
- 4 **ATmega microcontroller** The heart of your board.
- 5 **Analog in** Use these pins with `analogRead()`.
- 6 **GND and 5V pins** Use these pins to provide +5V power and ground to your circuits.
- 7 **Power connector** This is how you power your Arduino when it's not plugged into a USB port for power. Can accept voltages between 7-12V.
- 8 **TX and RX LEDs** These LEDs indicate communication between your Arduino and your computer. Expect them to flicker rapidly during sketch upload as well as during serial communication. Useful for debugging.
- 9 **USB port** Used for powering your Arduino UNO, uploading your sketches to your Arduino, and for communicating with your Arduino sketch (via `Serial.println()` etc.).
- 10 **Reset button** Resets the ATmega microcontroller.

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